OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

13.14 Combinations of elements and attributes for object types

13.14.1 barrier

A barrier is a continuous roadside object, which cannot be passed.

Table 108. Combinations of attributes and elements for `<object type="barrier">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<skeleton>` `<marking>` `<border>` `<material>` `<surface>` `<parkingSpace>`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<skeleton>` `<marking>` `<border>` `<material>` `<surface>` `<parkingSpace>`
barrier	hedge	common hedge made out of vegetation and bushes without gaps	not allowed	not recommended	recommended, size is defined in `<repeat>`	not allowed	optional, default value for @closed="false"	not allowed
	guardRail	metal guard rail along the side of the road (without the vertical poles)						
	jerseyBarrier	lower wall mostly made out of concrete to separate driving lanes						
	wall	higher wall out of concrete, bricks, stones …						
	railing	any kind of railing along the roadside						

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton> <marking> <border> <material> <surface> <parkingSpace>
	fence	metal or wooden fence						
	noiseProtec-tions	higher wall for noise protection						
	other	all other bar-rier objects subtypes that do not fit into current categories						

XML example

```
<object type="barrier"
        subtype="guardRail"
        name="guardRail"
        id="4000203"
        s="7.4615629425e+01"
        t="-1.4796332056e+01"
        zOffset="-0.328297280316"
        validLength="78.2913194973"
        orientation="none">
    <repeat s="7.4615629425e+01"
            length="2.2248725912e+00"
            distance="0.0000000000e+00"
            tStart="-1.4796332056e+01"
            tEnd="-1.4797490375e+01"
            widthStart="1.0000000000e-01"
            widthEnd="1.0000000000e-01"
            heightStart="3.0000000000e-01"
            heightEnd="3.0000000000e-01"
            zOffsetStart="-3.2829728032e-01"
            zOffsetEnd="-2.0316925494e-01" />
    <repeat s="7.6840502016e+01"
            length="1.9924939859e+00"
            distance="0.0000000000e+00"
            tStart="-1.4797490375e+01"
            tEnd="-1.4799675135e+01"
            widthStart="1.0000000000e-01"
            widthEnd="1.0000000000e-01"
            heightStart="3.0000000000e-01"
            heightEnd="3.0000000000e-01"
            zOffsetStart="-2.0316925494e-01"
            zOffsetEnd="-9.9028462309e-02" />
    <repeat s="7.8832996002e+01"
            length="3.3472807403e+00"
            distance="0.0000000000e+00"
            tStart="-1.4799675135e+01"
            tEnd="-1.4796827566e+01"
```

```
3               widthStart="1.0000000000e-01"
2               widthEnd="1.0000000000e-01"
4               heightStart="3.0000000000e-01"
2               heightEnd="3.0000000000e-01"
5               zOffsetStart="-9.9028462309e-02"
2               zOffsetEnd="5.4099928501e-02" />
6       <repeat s="8.2180276743e+01"
2               length="2.0283762167e+00"
7               distance="0.0000000000e+00"
2               tStart="-1.4796827566e+01"
8               tEnd="-1.4790365646e+01"
2               widthStart="1.0000000000e-01"
9               widthEnd="1.0000000000e-01"
3               heightStart="3.0000000000e-01"
0               heightEnd="3.0000000000e-01"
3               zOffsetStart="5.4099928501e-02"
1               zOffsetEnd="1.2659902980e-01" />
3       <repeat s="8.4208652959e+01"
2               length="8.1152058043e+00"
3               distance="0.0000000000e+00"
3               tStart="-1.4790365646e+01"
3               tEnd="-1.4802913672e+01"
4               widthStart="1.0000000000e-01"
3               widthEnd="1.0000000000e-01"
5               heightStart="3.0000000000e-01"
3               heightEnd="3.0000000000e-01"
6               zOffsetStart="1.2659902980e-01"
3               zOffsetEnd="1.7346901590e-01" />
7       <repeat s="9.2323858764e+01"
3               length="8.1137326328e+00"
8               distance="0.0000000000e+00"
3               tStart="-1.4802913672e+01"
9               tEnd="-1.4932009665e+01"
4               widthStart="1.0000000000e-01"
0               widthEnd="1.0000000000e-01"
4               heightStart="3.0000000000e-01"
1               heightEnd="3.0000000000e-01"
```

```
4               zOffsetStart="1.7346901590e-01"
2               zOffsetEnd="1.8665641186e-01" />
4       <repeat s="1.0043759140e+02"
3               length="8.1135361793e+00"
4               distance="0.0000000000e+00"
4               tStart="-1.4932009665e+01"
4               tEnd="-1.4813360696e+01"
5               widthStart="1.0000000000e-01"
4               widthEnd="1.0000000000e-01"
6               heightStart="3.0000000000e-01"
4               heightEnd="3.0000000000e-01"
7               zOffsetStart="1.8665641186e-01"
4               zOffsetEnd="1.7471050187e-01" />
8       <repeat s="1.0855112758e+02"
4               length="8.1144815334e+00"
9               distance="0.0000000000e+00"
5               tStart="-1.4813360696e+01"
0               tEnd="-1.4794688826e+01"
5               widthStart="1.0000000000e-01"
1               widthEnd="1.0000000000e-01"
5               heightStart="3.0000000000e-01"
2               heightEnd="3.0000000000e-01"
5               zOffsetStart="1.7471050187e-01"
3               zOffsetEnd="1.6260260750e-01" />
5        <repeat s="1.1666560911e+02"
4               length="8.1145989742e+00"
5               distance="0.0000000000e+00"
5               tStart="-1.4794688826e+01"
5               tEnd="-1.4789712834e+01"
6               widthStart="1.0000000000e-01"
5               widthEnd="1.0000000000e-01"
7               heightStart="3.0000000000e-01"
5               heightEnd="3.0000000000e-01"
8               zOffsetStart="1.6260260750e-01"
5               zOffsetEnd="1.5562310985e-01" />
9        <repeat s="1.2478020808e+02"
6               length="8.1144823549e+00"
```

```
            distance="0.0000000000e+00"
            tStart="-1.4789712834e+01"
            tEnd="-1.4785309018e+01"
            widthStart="1.0000000000e-01"
            widthEnd="1.0000000000e-01"
            heightStart="3.0000000000e-01"
            heightEnd="3.0000000000e-01"
            zOffsetStart="1.5562310985e-01"
            zOffsetEnd="1.4723730991e-01" />
      <repeat s="1.3289469044e+02"
            length="8.1147823670e+00"
            distance="0.0000000000e+00"
            tStart="-1.4785309018e+01"
            tEnd="-1.4796174945e+01"
            widthStart="1.0000000000e-01"
            widthEnd="1.0000000000e-01"
            heightStart="3.0000000000e-01"
            heightEnd="3.0000000000e-01"
            zOffsetStart="1.4723730991e-01"
            zOffsetEnd="1.7293727744e-01" />
      <repeat s="1.4100947281e+02"
            length="2.9774693217e+00"
            distance="0.0000000000e+00"
            tStart="-1.4796174945e+01"
            tEnd="-1.4801054064e+01"
            widthStart="1.0000000000e-01"
            widthEnd="1.0000000000e-01"
            heightStart="3.0000000000e-01"
            heightEnd="3.0000000000e-01"
            zOffsetStart="1.7293727744e-01"
            zOffsetEnd="1.9761954152e-01" />
</object>
```

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```

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2
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```

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3
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4
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```

13.14.2 building

A building is a closed object, which cannot be passed.

Table 109. Combinations of attributes and elements for <object type="building">

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton> <marking> <border> <material> <surface> <parkingSpace>
building	building	regular building like a house or office	optional	recommended	not allowed	optional	optional, default value for @closed="true"	not allowed
	busStop	bus stop with little roof and sign						
	tollBooth	small building with a barrier to collect tolls or charges						
	other	all other building objects subtypes that do not fit into current categories						

XML example

```
<object type="building"
        subType="building"
        name="house"
        id="0"
        s="2.4028125000000038e+01"
        t="1.2802136334240046e+01"
        zOffset="4.9999999999998934e-03"
        orientation="none"
        length="1.1300000000000001e+01"
        width="9.9900000000000002e+00"
        height="1.2230000000000000e+01"
        hdg="2.6413812899682183e+00"
        pitch="0.0000000000000000e+00"
        roll="0.0000000000000000e+00">
</object>
```

13.14.3 crosswalk

A crosswalk is an object on the road that can be passed. It is recommended to be defined as `<crossPath>` within a junction for pedestrian/bicycle simulation. If the crosswalk is defined as an object only, it will not be used for pedestrian/bicycle simulation.

Table 110. Combinations of attributes and elements for <object type="crosswalk">

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<marking> <surface> <material>	<skeleton> <border> <parkingSpace>
cross-walk	pedestrian	pedestrian crosswalk without zebra markings	not allowed	mandatory	not allowed	optional	optional, default value for @closed="true"	optional	not allowed
	bicycle	bicycle crossing, in Germany normally with red paint						optional	
	zebra	zebra crossing						optional	
	virtual	invisible crosswalk						not recommended	
	other	all other crosswalk objects subtypes that do not fit into current categories						optional	

XML example

```
<object type="crosswalk"
        id="10"
        s="10.0"
        t="0.0"
        zOffset="0.0"
        orientation="none"
        length="10.0"
        width="7.0"
        hdg="0.0"
        pitch="0.0"
        roll="0.0">
    <outlines>
        <outline id="0" closed="true">
            <cornerRoad s="5.0" t="3.5" dz="0.0" height="0.0" id="0"/>
            <cornerRoad s="8.0" t="-3.5" dz="0.0" height="0.0" id="1"/>
            <cornerRoad s="12.0" t="-3.5" dz="0.0" height="0.0" id="2"/>
            <cornerRoad s="15.0" t="3.5" dz="0.0" height="0.0" id="3"/>
        </outline>
    </outlines>
    <markings>
        <marking width="0.1"
                 color="white"
                 zOffset="0.005"
                 spaceLength ="0.05"
                 lineLength ="0.2"
                 startOffset="0.0"
                 stopOffset="0.0">
            <cornerReference id="0"/>
            <cornerReference id="1"/>
        </marking>
        <marking width="0.1"
                 color="white"
                 zOffset="0.005"
                 spaceLength ="0.05"
                 lineLength ="0.2"
                 startOffset="0.0"
```

```
                 stopOffset="0.0">
            <cornerReference id="2"/>
            <cornerReference id="3"/>
        </marking>
    </markings>
</object>
```

```
4
2
```

13.14.4 gantry

A gantry is an object above a road on which `<signals>` are placed.

Table 111. Combinations of attributes and elements for `<object type="gantry">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<skeleton>`	`<marking>` `<border>` `<material>` `<surface>` `<parkingSpace>`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton>	<marking> <border> <material> <surface> <parkingSpace>
gantry	gantry	has poles on either side of lanes and an overhead construction between them	not allowed	mandatory, for the entire gantry	not allowed	optional	not recommended, default value for @closed="true"	recommended, recommended to use <polyline>, of which first vertex and last vertex have @intersectionPoint	not allowed
	gantryHalf	has a pole on one side of the road and an overhead construction attached to it						recommended, recommended to use <polyline>, of which first vertex has an @intersectionPoint	
	other	all other gantry objects subtypes that do not fit into current categories						recommended	

XML example

```
<object type="gantry"
        subtype="gantry"
        name="SignGantry"
        id="4000001"
        s="25.0"
        t="-3.0"
        zOffset="0.00"
        roll="0"
        pitch="0"
        validLength=""
        orientation="none"
        height="5.5"
        length="0.5"
        width="6.5"
        dynamic="no"
        hdg="0">
    <skeleton>
        <polyline id="1">
            <vertexRoad s="25.0"
                        t="0.0"
                        dz="0.0"
                        width="0.5"
                        length="0.5"
                        id="0"
                        intersectionPoint="true" />
            <vertexRoad s="25.0"
                        t="0.0"
                        dz="5.25"
                        width="0.5"
                        length="0.5"
                        id="1" />
            <vertexRoad s="25.0"
                        t="-6.0"
                        dz="5.25"
                        width="0.5"
                        length="0.5"
```

```
                        id="2" />
            <vertexRoad s="25.0"
                        t="-6.0"
                        dz="0.0"
                        width="0.5"
                        length="0.5"
                        id="3"
                        intersectionPoint="true" />
        </polyline>
    </skeleton>
</object>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
```

13.14.5 obstacle

An obstacle is an object on or beside the road that cannot be passed.

Table 112. Combinations of attributes and elements for `<object type="obstacle">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<skeleton>` `<marking>` `<border>` `<material>` `<surface>` `<parkingSpace>`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton> <marking> <border> <material> <surface> <parkingSpace>
obstacle	advertisingColumn		either or		not allowed	optional	optional, default value for @closed="true"	not allowed
	art							
	seating							
	picknick							
	box							
	phonebooth							
	chargingStation							
	distributionBox	for example, electrical, communication						
	crashBox							
	dumpster							
	dustbin							
	fountain							
	gritContainer							

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton> <marking> <border> <material> <surface> <parkingSpace>
	hydrant							
	parkingMeter							
	pillar	for example, bridge pillars						
	plantPot							
	postBox							
	railing	for example, bicycle stand, handrail						
	rock							
	roadBlockage							
	wall							
	fence							
	other	all other obstacle objects subtypes that do not fit into current categories						

XML example

```xml
<object type="obstacle"
        subType="hydrant"
        name="GermanHydrant"
        id="1"
        s="8.3817187499999548e+01"
        t="-4.6359023698365620e+00"
        zOffset="0.0000000000000000e+00"
        orientation="none"
        radius="1.5000000000000000e-01"
        height="1.2520000000000000e+00"
        hdg="0.0000000000000000e+00"
        pitch="0.0000000000000000e+00"
        roll="0.0000000000000000e+00">
</object>
```

13.14.6 parkingSpace

A parkingSpace is an object on a lane on which vehicles are parked.

Table 113. Combinations of attributes and elements for `<object type="parkingSpace">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<parkingSpace> <material> <marking> <surface>	<skeleton> <border>
parkingSpace	openSpace	typically outdoors, no limit to the top	not allowed	mandatory	not allowed	optional	optional, default value for @closed="true"	optional	not allowed
	closedSpace	typically indoors, limit to the top for example, inside a building							
	other	all other parkingSpace objects subtypes that do not fit into current categories							

XML example

```
<object type="parkingSpace"
        subtype="open"
        name="parkingSpace_50deg"
        id="9"
        s="1.5e+01" t="-6.0e+00"
        zOffset="0.0"
        orientation="+"
        length="6.04e+00"
        width="2.5e+00"
        height="0.0"
        hdg="5.41052e+00"
        pitch="0.0"
        roll="0.0">
    <outlines>
        <outline id="51" closed="true">
            <cornerRoad s="11.60" t="-4.00" dz="0.001" height="0.0" id="0">
            <cornerRoad s="15.25" t="-8.15" dz="0.001" height="0.0" id="1">
            <cornerRoad s="18.39" t="-8.15" dz="0.001" height="0.0" id="2">
            <cornerRoad s="15.14" t="-4.00" dz="0.001" height="0.0" id="3">
        </outline>
    </outlines>
    <markings>
        <marking width="0.1"
                 color="white"
                 zOffset="0.005"
                 spaceLength="0.0"
                 lineLength="1.0"
                 startOffset="0.0"
                 stopOffset="0.0">
            <cornerReference id="0"/>
            <cornerReference id="1"/>
        </marking>
        <marking width="0.1"
                 color="white"
                 zOffset="0.005"
                 spaceLength="0.0"
```

```
                    lineLength="1.0"
                    startOffset="0.0"
                    stopOffset="0.0" >
                <cornerReference id="2"/>
                <cornerReference id="3"/>
            </marking>
        </markings>
        <parkingSpace access="all">
        </parkingSpace>
    </object>
```

```
4
2
4
3
4
4
4
5
4
6
```

```
<object type="parkingSpace"
        subtype="closed"
        name="parkingGarage"
        id="10"
        s="3.0e+01"
        t="-6.0"
        zOffset="0.0"
        length="6.0"
        width="3.45"
        height="2.64"
        hdg="0.0"
        roll="0.0"
        pitch="0.0"
        orientation="+">
    <outlines>
        <outline id="53" fillType="concrete" outer="false">
            <cornerLocal v="-3.0" u="-1.43" z="0.0" height="2.11" id="0">
            <cornerLocal v="-3.0" u="1.43" z="0.0" height="2.11" id="1">
            <cornerLocal v="2.705" u="1.43" z="0.0" height="2.11" id="2">
            <cornerLocal v="2.705" u="-1.43" z="0.0" height="2.11" id="3">
        </outline>
    </outlines>
    <parkingSpace access="residents">
    </parkingSpace>
</object>
```

```
3
2
4
2
5
```

13.14.7 pole

A pole is thin long object beside drivable lanes.

Table 114. Combinations of attributes and elements for `<object type="pole">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<skeleton>`	`<border>` `<parkingSpace>` `<material>` `<marking>` `<surface>`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton>	<border> <parkingSpace> <material> <marking> <surface>
pole	emergencyCallBox		either or		not allowed	optional	optional, default value for @closed="true"	optional	not allowed
	permanentDelineator							optional	
	bollard							optional	
	trafficSign	pole for Traffic Signs						optional	
	trafficLight	pole for trafficLight and trafficSign objects						recommended if it has an extrusion. Use `<polyline>`, of which first vertex has an @intersectionPoint.	
	powerPole	has power cables attached						optional	

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton>	<border> <parkingSpace> <material> <marking> <surface>
	street-Lamp	has a light source. Might also have traf-ficSigns or traffi-cLights at-tached to it.						recommended if it has an arm. Use <polyline>, of which first vertex has an @intersectionPoint.	
	windTur-bine							optional	
	other	all other pole ob-jects sub-types that do not fit into current categories						optional	

XML example

```
<object type="pole"
        subtype="emergencyCallBox"
        name="emergencyCallBoxGerman"
        id="4"
        s="1.1350690873e+02"
        t="-5.4331497312e+00"
        zOffset="0.000"
        validLength="0.0"
        hdg="0.000"
        orientation="none"
        height="1.800"
        radius="0.05"
        roll="0"
        pitch="0"
        dynamic="no">
</object>
```

13.14.8 roadMark

A roadMark object is painted on the road and can be passed.

Table 115. Combinations of attributes and elements for `<object type="roadMark">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<marking> <material> <surface>	<skeleton> <border> <parkingSpace>
road-Mark	arrowLeft		not allowed	mandatory	not allowed	optional	optional, de-fault value for @closed="true"	optional	not allowed
	arrowLeftLeft								
	arrowLeft-Right								
	arrowRight								
	ar-rowRightRight								
	ar-rowRightLeft								
	arrowStraight								
	arrow-StraightLeft								
	arrow-StraightRight								
	arrow-StraightLeftRight								
	ar-rowMergeLeft								
	arrowMerg-eRight								

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<marking> <material> <surface>	<skeleton> <border> <parkingSpace>
	signalLines	these are referenced by a signal							
	text	for example, YIELD or 50, might be referenced by a signal							
	symbol	for example, Wheelchair or bicycle							
	paint								
	area	for example, restricted area, keep clear area							

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<marking> <material> <surface>	<skeleton> <border> <parkingSpace>
	other	all other roadMark objects subtypes that do not fit into current categories							

XML example

```
<object type="roadMark"
        subtype="arrowStraight"
        name="arrowStraightWhite"
        id="5"
        s="1.5856415945e+00"
        t="1.7853615772e+00"
        zOffset="0.000"
        orientation="none"
        hdg="0.000"
        length="7.373"
        width="0.552"
        roll="0"
        pitch="0"
        validLength="0"
        height="0"
        dynamic="no">
    <outline id="1" outer="true" closed="true" laneType="driving">
        <cornerLocal u="-3.6386" v="0.1123" z="0.0000" height="0.0000" id="0"/>
        <cornerLocal u="-3.6864" v="-0.1213" z="0.0000" height="0.0000" id="1"/>
        <cornerLocal u="0.8244" v="-0.0792" z="0.0000" height="0.0000" id="2"/>
        <cornerLocal u="0.8167" v="-0.2762" z="0.0000" height="0.0000" id="3"/>
        <cornerLocal u="3.6864" v="-0.0093" z="0.0000" height="0.0000" id="4"/>
        <cornerLocal u="0.8104" v="0.2762" z="0.0000" height="0.0000" id="5"/>
        <cornerLocal u="0.7872" v="0.0582" z="0.0000" height="0.0000" id="6"/>
        <cornerLocal u="-3.6386" v="0.1123" z="0.0000" height="0.0000" id="7"/>
    </outline>
    <material roadMarkColor="white"/>
    <validity fromLane="1" toLane="1"/>
</object>
```

```
3
2
4
2
5
2
6
2
7
2
8
2
9
```

13.14.9 roadSurface

A roadSurface object is on the road and can be passed.

Table 116. Combinations of attributes and elements for `<object type="roadSurface">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<material>` `<surface>`	`<skeleton>` `<border>` `<parkingSpace>` `<marking>`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<material> <surface>	<skeleton> <border> <parkingSpace> <marking>
roadSurface	manhole	mostly metal cover to access sewerage tunnels	either or		not allowed	optional	optional, default value for @closed="true"	optional	not allowed
	pothole	road damage							
	patch	road damage that has been fixed							
	speedbump	mostly raised surface to prevent higher speeds							
	drainGutter	water drainage							
	other	all other roadSurface objects subtypes that do not fit into current categories							

XML example

```
<object type="roadSurfaceElement"
        subType="patch"
        name="Rd_Damage_Patch_22_CRG"
        id="2"
        s="3.1064163564293011e+01"
        t="1.6886219784199805e+00"
        zOffset="0.0000000000000000e+00"
        validLength="0.0000000000000000e+00"
        orientation="none"
        length="1.9179999999999999e+00"
        width="3.2229999999999999e+00"
        height="0.0000000000000000e+00"
        hdg="5.7401170550235827e+00"
        pitch="0.0000000000000000e+00"
        roll="0.0000000000000000e+00">
    <surface>
        <CRG file="Rd_Damage_Patch_22_Center.crg" hideRoadSurfaceCRG="true" zScale="1"/>
    </surface>
</object>
```

13.14.10 trafficIsland

A trafficIsland object is on the road and should not be passed by vehicles.

Table 117. Combinations of attributes and elements for `<object type="trafficIsland">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<material>` `<border>` `<marking>`	`<skeleton>` `<surface>` `<parkingSpace>`
trafficIs-land	island	typical traf-fic island with some curbstone, road mark	either or		not allowed	not recommended	recommended, default value for @closed="true"	optional	not allowed
	other	all other trafficIsland objects sub-types that do not fit into current categories							

XML example

```
<object type="trafficIsland"
        subtype="island"
        name="ExampleIsland"
        id="8"
        s="5.0000000000000441e+01"
        t="4.4053649617126212e-13"
        zOffset="0.0000000000000000e+00"
        orientation="none"
        length="5.0000000000000000e+00"
        width="1.0000000000000000e+00"
        height="1.0000000000000000e-01"
        hdg="0.0000000000000000e+00"
        pitch="0.0000000000000000e+00"
        roll="0.0000000000000000e+00">
    <outlines>
        <outline id="50" fillType="cobble" closed="true">
            <cornerRoad s="52.5" t="1.5" dz="0.0" height="0.1"/>
            <cornerRoad s="52.6" t="1.1" dz="0.0" height="0.1"/>
            <cornerRoad s="52.7" t="0.7" dz="0.0" height="0.1"/>
            <cornerRoad s="52.8" t="0.6" dz="0.0" height="0.1"/>
            <cornerRoad s="52.9" t="0.55" dz="0.0" height="0.1"/>
            <cornerRoad s="53.0" t="0.5" dz="0.0" height="0.1"/>
            <cornerRoad s="57.0" t="0.5" dz="0.0" height="0.1"/>
            <cornerRoad s="57.5" t="0.5" dz="0.0" height="0.01"/>
            <cornerRoad s="61.5" t="0.5" dz="0.0" height="0.01"/>
            <cornerRoad s="62.0" t="0.5" dz="0.0" height="0.1"/>
            <cornerRoad s="66.0" t="0.5" dz="0.0" height="0.1"/>
            <cornerRoad s="66.1" t="0.55" dz="0.0" height="0.1"/>
            <cornerRoad s="66.2" t="0.6" dz="0.0" height="0.1"/>
            <cornerRoad s="66.3" t="0.7" dz="0.0" height="0.1"/>
            <cornerRoad s="66.4" t="1.1" dz="0.0" height="0.1"/>
            <cornerRoad s="66.5" t="1.5" dz="0.0" height="0.1"/>
            <cornerRoad s="66.4" t="1.9" dz="0.0" height="0.1"/>
            <cornerRoad s="66.3" t="2.3" dz="0.0" height="0.1"/>
            <cornerRoad s="66.2" t="2.4" dz="0.0" height="0.1"/>
            <cornerRoad s="66.1" t="2.45" dz="0.0" height="0.1"/>
```

```
            <cornerRoad s="66.0" t="2.5" dz="0.0" height="0.1"/>
            <cornerRoad s="62.0" t="2.5" dz="0.0" height="0.1"/>
            <cornerRoad s="61.5" t="2.5" dz="0.0" height="0.01"/>
            <cornerRoad s="57.5" t="2.5" dz="0.0" height="0.01"/>
            <cornerRoad s="57.0" t="2.5" dz="0.0" height="0.1"/>
            <cornerRoad s="53.0" t="2.5" dz="0.0" height="0.1"/>
            <cornerRoad s="52.9" t="2.45" dz="0.0" height="0.1"/>
            <cornerRoad s="52.8" t="2.4" dz="0.0" height="0.1"/>
            <cornerRoad s="52.7" t="2.3" dz="0.0" height="0.1"/>
            <cornerRoad s="52.6" t="1.9" dz="0.0" height="0.1"/>
        </outline>
    </outlines>
    <borders>
        <border width="0.1" type="curb" outlineId="50" useCompleteOutline="true"/>
    </borders>
</object>
```

```
4
2
4
3
4
4
4
5
4
6
4
7
4
8
4
9
5
0
5
1
5
2
```

13.14.11 tree

A tree object is a single vegetational object with a trunk.

Table 118. Combinations of attributes and elements for `<object type="tree">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	`<repeat>` with @distance="0"	`<repeat>` with @distance>"0"	`<outlines>`	`<skeleton>`	`<material>` `<surface>` `<border>` `<parkingSpace>` `<marking>`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton>	<material> <surface> <border> <parkingSpace> <marking>
tree	needle	needle tree	either or		not allowed	optional	recommended, 2 entries, one for trunk, one for crown		not allowed
	leaf	leaf tree							
	palm	palm tree							
	other	all other tree objects sub-types that do not fit into current categories							

XML example

```xml
<object type="tree"
        subtype="leaf"
        name="leafTree"
        id="6"
        s="9.3817187499999548e+01"
        t="-5.001023698365620e+00"
        zOffset="-1.00"
        roll="0"
        pitch="0"
        validLength=""
        orientation="none"
        height="7.50"
        length="4.00"
        width="4.00"
        dynamic="no"
        hdg="0">
    <skeleton>
        <polyline id="1">
            <vertexLocal u="-0.2" v="1.0" z="1.120" radius="0.15" id="0" intersectionPoint="true" />
            <vertexLocal u="-0.2" v="1.0" z="4.500" radius="0.12" id="1"/>
        </polyline>
    </skeleton>
    <outlines>
        <outline id="2" closed="true">
            <cornerLocal u="2.0" v="0.0" z="4.0" height="3.5" id="0"/>
            <cornerLocal u="1.0" v="2.0" z="4.0" height="3.5" id="1"/>
            <cornerLocal u="-1.0" v="2.0" z="4.0" height="3.5" id="2"/>
            <cornerLocal u="-2.0" v="0.0" z="4.0" height="3.5" id="3"/>
            <cornerLocal u="-1.0" v="-2.0" z="4.0" height="3.5" id="4"/>
            <cornerLocal u="-1.0" v="-2.0" z="4.0" height="3.5" id="5"/>
        <outline>
    </outlines>
</object>
```

```
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0
3
1
3
2
3
3
```

13.14.12 vegetation

A vegetation object is a single vegetational object without a trunk or an area of vegetation.

Table 119. Combinations of attributes and elements for `<object type="vegetation">`

@type	@subType	Description	Boundingbox using @radius and @height	Boundingbox using @width, @length and @height	<repeat> with @distance="0"	<repeat> with @distance>"0"	<outlines>	<skeleton> <material> <surface> <border> <parkingSpace> <marking>
vegetation	bush	a single bush	either or		not allowed	optional	optional, default value for @closed="true"	not allowed
	forest	an area that is a forest						
	hedge	a single hedge						
	other	all other vegetation objects subtypes that do not fit into current categories						

XML example

```
<object type="vegetation"
        subType="bush"
        name="VegBush06"
        id="3"
        s="5.5223437499999534e+01"
        t="1.1123800966684282e+01"
        zOffset="-1.6500000000000004e-01"
        validLength="0.0000000000000000e+00"
        orientation="none"
        radius="1.5409999999999998e+00"
        height="3.1600000000000001e+00"
        hdg="0.0000000000000000e+00"
        pitch="0.0000000000000000e+00"
        roll="0.0000000000000000e+00">
</object>
```